

Boyu Cai

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Education

ShanghaiTech University & Institute of Automation, Chinese Academy of Sciences, *Joint MS in Computer Science* *Sept 2024 – June 2027*

- **Advisors:** [Weiming Hu](#), [Yan Xu](#), and [Li Yang](#).
- Research Interest: Embodied Intelligence, Vision Language Model

Dalian Maritime University, *BS in Software Engineering* *Sept 2020 – June 2024*

- GPA: 4.01/5.00, Rank: 2/93
- COSCO Shipping Excellence Navigation Scholarship (**Top 2%**), etc

Publications

LANCE: Learning What to See in Vision-Language-Action Policies via Language-Anchored Action Queries, **First Author** Under Review

- Identified **language de-anchoring** in VLA policies: action gener can rely on visual shortcuts rather than instruction semantics.
- Proposed Language-Anchored Action Queries with KL-based prior-posterior alignment.
- Introduced a sampling-based language-action loss to prevent prior collapse and learn action-discriminative language representations.
- Achieved **90.6%** success on LIBERO-Plus Lang and **63%** on VLA-Arena Unseen Object, with real-robot validation demonstrating the ability to **compose multiple atomic subtasks at test time**.

Dynamic-Robust Photometric-Semantic Reconstruction for Open-Vocabulary 3D Scene Understanding, **First Author** ECCV 2026

- Proposed **SPAR**, a feed-forward dynamic-robust semantic-photometric reconstruction framework from sparse, unposed, and dynamic observations.
- Photometric and open-vocabulary semantic information are jointly modeled in a unified latent scene representation **without explicit 3D geometric supervision**.
- Introduced reconstruction-supervised dynamic-region modeling **without ground-truth motion masks or explicit 3D supervision**.
- Achieved **88.5% motion-mask mIoU** and state-of-the-art NVS performance.

SRA-Det: Learning Omni-Grained Open-Vocabulary Detection Beyond Category Names, **First Author** CVPR 2026

- Proposed **SRA-Det**, which retrieves multiple semantic cues from token-level text features for omni-grained open-vocabulary detection.
- Built an automatic attribute-augmentation pipeline using LLM-generated visual attributes and CLIP-based instance-level filtering.
- Improved fine-grained detection from **41.1** to **54.9 (+13.8)** while preserving performance on standard category-name detection.

Focus More on What? Guiding Multi-Task Training for End-to-End Person Search, **First Author** TCSVT 2025

- Systematically studied joint training of detection and re-identification in end-to-end person search, revealing **imbalanced and conflicting optimization** between the two tasks.
- Proposed a **Guiding Multi-Task Harmonious Learning** module that uses intra- and cross-task feature interaction to reduce feature conflicts and improve task-specific representations.
- Designed an iterative multi-stage training strategy to better balance detection and re-identification, further alleviating conflicts during joint optimization.

Temporal Representation Enhancement (TRE): Learning to Forget Dominant Patterns for More Discriminative Spiking Features CVPR 2026

- Proposed **Temporal Representation Enhancement (TRE)** to suppress dominant temporal patterns and learn more complementary features across time steps.
- Improved temporal feature discrimination and multi-timestep learning in Spiking Neural Networks.

Towards More Discriminative Feature Learning in SNNs with Temporal-Self-Erasing Supervision AAAI 2025

- Proposed **Temporal-Self-Erasing (TSE)** supervision to dynamically suppress dominant regions across time steps.
- Encouraged complementary temporal representations and improved multi-timestep inference in Spiking Neural Networks.

Research Experience

CyboPal, *Research Intern*

Apr. 2026 – Jul. 2026

- Studied semantic instruction following and OOD generalization in VLA models, and developed the language-anchored action-query method described in LANCE.
- Achieved **90.6%** success on LIBERO-Plus Lang and **63%** on VLA-Arena Unseen Object, and built a dual-arm real-robot benchmark for complex manipulation tasks.
- Contributed to robot-client testing and the adaptation of teleoperation and data-collection pipelines.

Institute of Automation, Chinese Academy of Sciences (CASIA), *Research Intern*

Dec. 2023 – Sep. 2024

- Conducted open-vocabulary detection research in a CASIA–Alibaba project, including model development, data construction, and migration to **MMDetection**.
- Developed hard-negative category sampling, improving LVIS AP from **32.8** to **36.2**; multi-source training further achieved **46.0 AP**.

Projects

DDSS: Dynamic DDoS Traffic Scrubbing System

[P4-DDOS](#) 

- Built a dynamic DDoS traffic scrubbing system using P4-programmable switches, VNFs, and SDN-based rule deployment.
- Reduced network latency during dynamic attacks while introducing minimal CPU overhead.

Skills

Languages: Chinese, Japanese, Familiar with English reading and writing

Technologies: Python, PyTorch, Linux